Isaiah Thompson Ocansey, Ph.D.

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EXPERIENCE

Graduate Research Assistant, Machine Learning and Quantitative Modeling 01/2022 – 12/2025
University of Texas at El Paso, Department of Mathematical Sciences *El Paso, TX*

- Engineered LogTinyLLM, a compact transformer delivering 98.83% anomaly-detection accuracy at 70% lower compute cost than large-LLM baselines through LoRA fine-tuning; first-author paper under review at Annals of Data Science (arXiv:2507.11071).
- Designed a cross-silo federated learning pipeline with differential privacy, enabling institutions to train shared models on protected records without centralizing data; second paper under review.
- Segmented 5,960 home equity borrowers across 13 risk variables using K-Means and hierarchical clustering; analysis projected a 35% reduction in credit default rates and a 15% lift in loan approval accuracy.
- Accelerated experiment turnaround on multi-modal deep learning workloads by implementing distributed PyTorch training with mixed precision and FSDP sharding on TACC supercomputing infrastructure.

Data Analytics Consultant, Strategic Decision Support 05/2023 – 08/2024
University of Texas at El Paso, Graduate School *El Paso, TX*

- Deployed student success forecasting models at 94% accuracy using ensemble and deep learning methods, equipping advisers with ranked at-risk cohorts early enough to intervene.
- Advised the Board of Regents and senior leadership on enrollment policy and multimillion-dollar capital decisions; institutional enrollment rose 15%.
- Cut ad-hoc reporting requests 40% by building self-service Power BI dashboards consolidating 10+ institutional KPIs.

Machine Learning and Data Engineering Intern 06/2025 – 08/2025
University of Texas System *Austin, TX*

- Delivered real-time operational risk visibility to executives across 14 institutions by integrating 15+ ITSM data sources through REST APIs and Power Automate into Power BI.
- Eliminated 15+ hours of manual processing weekly and improved audit readiness 25% by automating compliance tracking and audit reporting pipelines.
- Identified $2M+ in federal AI grant opportunities by conducting a structured funding landscape analysis across federal grant databases, shaping system-wide AI investment priorities.

Technical Specialist, Quantitative Risk and Data Analytics 08/2018 – 12/2021
AT&T Inc. *Prague, Czech Republic*

- Lifted Net Promoter Score 20% and retained $1.2M in annual revenue by leading customer experience analytics that isolated and resolved systemic service failures.
- Directed a cross-functional team of 8 analysts deploying statistical process control across network operations, reducing service disruptions 15%.
- Built large-scale SQL extraction and transformation pipelines powering enterprise KPI tracking and executive reporting.

PROJECTS

The Label Budget github.com/ocansey/label-budget

- Quantifies how many labels must be purchased before a supervised detector outperforms an unsupervised model requiring none; Isolation Forest, gradient boosting, and RBF-SVM built from first principles in approximately 500 lines, no libraries.
- Demonstrated that allowing the unsupervised model to select which flows reach a human analyst halves the label budget required for equivalent detection quality, a direct reduction in analyst hours at identical performance.
- Shipped with 36 tests and a documented limitations section defining where the finding does not generalize; runs entirely client-side.

Drift Arena github.com/ocansey/drift-arena

- Live champion/challenger console for card-fraud detection that surfaces silent model decay: shadow scoring across 100% of traffic, PSI drift monitoring on a rolling window, adaptive traffic reallocation on alarm, and McNemar-gated promotion on discordant pairs.
- Reproduces a fraud model whose recall collapses from 97% to 25% under an attack profile outside its feature set, then demonstrates the monitoring machinery that detects and corrects it; all statistics computed live.

PUBLICATIONS

- Complete record: scholar.google.com/citations?user=pen_i60AAAAJ&hl=en (10 publications)
- Ocansey, I.T., Bhattacharya, R., & Sen, T. (2025). LogTinyLLM: Contextual Log Anomaly Detection Using Tiny Large Language Models. Under review, IEEE Transactions on Neural Networks and Learning Systems. arXiv:2507.11071.
- Ocansey, I.T., Sen, T., & Bhattacharya, R. (2026). DP-FLogTinyLLM: Differentially Private Federated Log Anomaly Detection Using Tiny LLMs. Under review.

- Ocansey, I.T., & Yeboah, M.M. (2026). AI-Driven Privacy-Preserving Techniques in US Healthcare Cybersecurity. Magna Scientia Advanced Research and Reviews.
- Ntiakoh, A., Ocansey, I.T., & Amoakoh, C. (2026). The Role of Artificial Intelligence in Reducing Financial Fraud in U.S. Healthcare Systems. Magna Scientia Advanced Research and Reviews.
- Ocansey, I.T. (2025). Comparative Analysis of Hierarchical and K-Means Clustering Algorithms on Home Equity Loan Data. Under review, Journal of Statistical Computation and Simulation.

SKILLS

Machine Learning and AI: Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Generative AI, Anomaly Detection, Predictive Modeling, Federated Learning, Model Compression, LoRA Fine-Tuning, Instruction Tuning, Retrieval-Augmented Generation (RAG)
Statistics and Quantitative Methods: Statistical Inference, Survival Analysis, Time Series, Hypothesis Testing, Experimental Design, Credit Risk Modeling, Statistical Process Control
Programming and Data Engineering: Python, SQL, R, Bash, LaTeX, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Power BI, Power Automate, REST APIs
MLOps and Infrastructure: MLOps, Model Monitoring, Distributed Training, FSDP, Mixed Precision, Gradient Checkpointing, AWS, Microsoft Azure, Azure ML, Apache Spark, Docker, Git, Linux, Slurm/HPC (TACC)

AWARDS & HONORS

- UTEP Graduate Research Fellowship: $75,000 total award, 2022-2026
- Member, American Statistical Association
- Affiliate, Texas Advanced Computing Center (TACC)
- AWS Certified Machine Learning Engineer (in progress)

EDUCATION

University of Texas at El Paso
Ph.D., Data Science. Dissertation: Multi-Modal Anomaly Detection Using Deep Learning Techniques. Advisor: Dr. Ritwik Bhattacharya. *05/2026*

University of Silesia, Poland
M.S., Mathematics of Finance and Economics. Thesis: Affine Interest Rate Models, term structure, no-arbitrage bond pricing, yield curve dynamics. *09/2017*

University of L'Aquila, Italy
M.S., Engineering Mathematics. *09/2017*

University for Development Studies, Ghana
B.S., Financial Mathematics. *04/2015*